

IntelliTrack: The Approach to Leveraging Machine Learning for Indoor room prediction Using WiFi Signals

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ABSTRACT

The increasing complexity of indoor environments, such as hospitals, universities, and public buildings, presents a challenge for traditional GPS-based navigation systems due to signal attenuation and the inability to determine vertical positioning accurately. This study introduces IntelliTrack, a novel approach utilizing WiFi fingerprinting to address these limitations by leveraging existing WiFi infrastructure for indoor positioning. By collecting and analyzing WiFi signal strength data from various endpoints within a university setting, and correlating them with specific room numbers, we developed a machine learning model capable of predicting a user's precise location inside a building. Our findings indicate that WiFi fingerprinting, combined with advanced machine learning techniques, offers a viable solution for enhancing navigation in indoor environments where GPS signals are unreliable. This paper outlines our methodology, from data collection and preprocessing to model training and validation, and discusses the potential applications and implications of this technology in real-world settings.

KEYWORDS

WiFi Fingerprinting, Machine Learning Algorithms, Signal Strength Analysis, Deep Learning for IPS, Advanced Indoor Mapping, Predictive Location Modeling, Indoor GPS Alternatives, Data-Driven Positioning, Signal Processing for Navigation, Smart Building Navigation

1 INTRODUCTION

Indoor Positioning Systems (IPS) have emerged as an essential innovation for navigating complex indoor spaces, finding applications across emergency services, retail analytics, and everyday navigation within large buildings like hospitals and universities. Traditional Global Positioning System (GPS) solutions, while effective outdoors, significantly underperform in indoor environments due to signal loss and the inability to accurately discern vertical positions, such as different floors within a building [4]. This challenge has spurred the exploration of alternative technologies for indoor navigation, with WiFi fingerprinting emerging as a promising solution [1, 5].

IntelliTrack represents a significant leap forward in this domain by harnessing the ubiquity of WiFi infrastructure to facilitate precise indoor location tracking. Our approach involved the development of a machine learning model trained on a dataset of WiFi signal strengths collected from multiple access points across our university campus. Each data point was meticulously labeled with corresponding room numbers to enable the prediction of a user's

location based on the WiFi signals received by their mobile device. This paper details the design and implementation of IntelliTrack, including the underlying machine learning algorithms that enable WiFi signal analysis, and presents a comprehensive evaluation of the system's efficacy in diverse indoor settings. Our work demonstrates the potential of WiFi fingerprinting to overcome the limitations of GPS in indoor environments, offering a new avenue for accurate and efficient indoor navigation.¹

RELATED WORK

Indoor positioning systems (IPS) have rapidly evolved, moving from traditional methods to more sophisticated techniques that leverage machine learning and deep learning for enhanced accuracy and reliability. The proliferation of indoor positioning applications across various domains necessitates the exploration of new trends and advancements that significantly contribute to the field.

Traditional Methods. Traditionally, IPS relied heavily on methods such as Received Signal Strength Indication (RSSI) and the deployment of Wi-Fi access points [1]. For instance, Abidin et al. [1] and Zeng et al. [6] demonstrated the use of RSSI fingerprints for positioning, where the latter improved accuracy through Gaussian filtering and Bayesian algorithms. Moreover, Zhu [7] optimized the KNN algorithm for indoor localization, addressing computational and efficiency challenges.

Machine Learning Approaches. The integration of machine learning algorithms has marked a significant transition in IPS development. Techniques such as Support Vector Machines (SVM), k-Nearest Neighbors (kNN), and decision trees have been explored for their potential in improving positioning accuracy with complex indoor environments [1]. These approaches attempt to model the inherently noisy and reflective indoor spaces more accurately than traditional methods.

Deep Learning and New Trends. The advent of deep learning has introduced a new paradigm in IPS, offering promising results in dealing with the complexities of indoor environments. Wietrzykowski [5] utilized linear-chain Conditional Random Fields to infer user positions, harnessing the power of sequence prediction models for

¹In the development of IntelliTrack, we utilized advanced Language Models (LMs), specifically designed for natural language understanding and text generation, to assist in various stages of our research. These models contributed significantly to the analysis of literature, helping us synthesize vast amounts of data and insights from related works efficiently. Furthermore, LMs supported the generation of coherent and comprehensive narratives within our paper, ensuring a robust articulation of concepts, methodologies, and discussions.

trajectory consistency. Deep learning methods, particularly Deep Neural Networks (DNNs), have been applied for fingerprint-based positioning, showing remarkable improvements in accuracy and robustness over traditional machine learning methods [1].

Furthermore, recent studies have explored the utilization of Wi-Fi signals combined with inertial measurement units (IMUs) and environmental sensors, offering a multi-modal approach to IPS. This integration enhances the system’s understanding of indoor spaces, enabling more precise and context-aware positioning [2].

Conclusion. The exploration of new trends in IPS, particularly the application of deep learning and multi-modal data integration, presents a promising avenue for research and development. These advancements not only address the accuracy and efficiency challenges but also open up new possibilities for indoor positioning applications across various sectors.

These studies collectively highlight the diverse methodologies and innovations being explored in the realm of indoor positioning systems. From advanced signal processing and probabilistic inference to the application of machine learning algorithms, the field is witnessing a broad array of strategies aimed at enhancing accuracy and reliability.

2 METHODOLOGY

This section delineates the methodological framework employed in the IntelliTrack project, aimed at leveraging WiFi signal strength for precise indoor positioning. The approach is systematically divided into data collection, preprocessing, and machine learning model development phases, each underpinned by rigorous data analysis and computational methodologies.

2.1 Data Collection

Data collection was facilitated by a specialized function within our Android application, meticulously engineered to capture a comprehensive suite of WiFi signal characteristics. These included the signal strength (dBm), Service Set Identifier (SSID), Basic Service Set Identifier (BSSID), signal level, frequency of the signal, estimated distance to the access point (though this was often null and thus not always applicable), channel width, and center frequencies (centerFreq0 and centerFreq1) for dual-band signals. The application was deployed on Android devices due to their capability to perform WiFi scans at fixed intervals of 30 seconds, a feature restricted on iOS platforms. Collection efforts were conducted within multiple indoor environments and adjacent outdoor areas to capture a broad spectrum of signal variability. Movement patterns during collection were randomized to mitigate bias and ensure a representative dataset. This methodological choice reflects the necessity to navigate platform limitations while capturing data reflective of diverse and dynamic signal landscapes.

2.2 Data Preprocessing

Given the raw nature of the collected WiFi signal data, a comprehensive preprocessing pipeline was established to refine the dataset for analytical use. This included:

Normalization and Anomaly Filtering: In this initial stage of data preprocessing, WiFi signal strengths, inherently variable due

to factors like device discrepancies and environmental conditions, were standardized to a uniform metric scale. This normalization process facilitates comparison across different data points and devices, enhancing the reliability of subsequent analyses. Concurrently, the dataset underwent a rigorous filtering process to identify and remove anomalous readings—outliers that could skew the model’s learning process. Such anomalies could arise from technical glitches, extreme environmental interference, or data entry errors. Through the analysis of data distribution across various network frequencies, we determined that exclusively utilizing 5GHz networks results in more consistent data and a better differentiation between distinct rooms. This observation underscores the superior performance of 5GHz networks in our indoor positioning system, offering a more reliable basis for distinguishing spatial locations.

Feature Extraction: The raw WiFi signal data presents a rich but overwhelming array of information. The feature extraction process involved distilling this information down to the most pertinent attributes with a direct bearing on positioning accuracy. Key features such as the amplitude of signal strength (measured in dBm) and the signal’s frequency (measured in MHz) were prioritized. These features are crucial because they encapsulate the essence of the signal’s interaction with the indoor environment—factors like distance from the access point and interference from physical barriers. Extracting and focusing on these salient features allows the machine learning model to more accurately correlate specific signal patterns with physical locations, thus enhancing its predictive precision.

Dimensionality Reduction: High-dimensional datasets, while rich in information, pose significant challenges for machine learning models in terms of computational complexity and the risk of overfitting. To mitigate these issues, a dimensionality reduction strategy was implemented. This process involves selecting and retaining only the most informative signals—those that contribute most significantly to the model’s ability to discern distinct locations. By prioritizing signals based on their strength and eliminating redundant or less informative attributes, the dimensionality reduction process simplifies the model’s learning environment. This not only accelerates the training phase but also improves the model’s ability to generalize from the training data to new, unseen environments, ultimately enhancing the accuracy and reliability of the indoor positioning system.

2.3 Machine Learning Model Development

The development of IntelliTrack’s predictive model was underpinned by an exhaustive exploration of several advanced machine learning algorithms, aimed at identifying the most effective method for predicting indoor locations from WiFi signal fingerprints. This iterative process involved the evaluation of various algorithms, including but not limited to Gradient Boosting, to ascertain their suitability based on accuracy, computational efficiency, and the complexity of the indoor positioning problem. Below we detail the structured approach taken in this exploration:

Algorithm Evaluation: An extensive review of machine learning

algorithms was conducted, encompassing Gradient Boosting (XGBoost), Random Forest, Support Vector Machines, and Neural Networks. Each algorithm was subjected to a preliminary evaluation to assess its potential in accurately modeling the complex relationship between WiFi signal characteristics and spatial locations. The criteria for this evaluation included model accuracy, interpretability, and computational requirements, ensuring a comprehensive assessment of each algorithm’s strengths and limitations.

Model Training and Validation: Following the selection process, multiple models were trained using a carefully partitioned dataset, divided into training and validation sets. This approach facilitated not only the optimization of each model’s parameters but also allowed for an unbiased evaluation of their performance. Cross-validation techniques were employed across all models to enhance the robustness of the training process and ensure the generalizability of the predictive models.

Feature Engineering and Selection: Integral to the model development phase was a meticulous process of feature engineering, which involved the creation, evaluation, and selection of features based on their predictive value. This process was iteratively refined, leveraging the insights gained from initial model evaluations to enhance the predictive accuracy of the algorithms.

Performance Evaluation and Algorithm Selection: Each algorithm was rigorously assessed through a series of performance metrics, including accuracy (measured as the mean distance error from the actual location), precision, recall, and the F1 score. This comprehensive evaluation facilitated a data-driven decision-making process, ultimately leading to the selection of the Gradient Boosting algorithm for its superior performance in modeling the intricacies of indoor positioning using WiFi signals. Chapter 3 shows the results in detail.

Implementation and Optimization: The final model was implemented using the xgboost library, with a concerted focus on parameter optimization using GridSearch to strike an optimal balance between training efficiency and predictive accuracy. This phase also addressed challenges related to data imbalance and the high-dimensionality of the feature space, ensuring the model’s effectiveness across various indoor environments.

3 RESULTS

This section presents the findings of our study, focusing on the performance evaluation of different machine learning models developed to address our localization challenge. We detail the evaluation metrics employed to assess model performance, followed by a concise analysis of the results obtained from each model. The models considered in this study include Decision Trees, Random Forest Classifiers, and XGBoost. Through comparative analysis, we aim to identify the most effective model based on our evaluation criteria.

3.1 Evaluation Metrics

To evaluate the performance of our ML models and ensure reliability and validity of results, we utilized two primary evaluation strategies: a train-test split and k-fold cross-validation. The train-test split method provided us with an initial assessment of model performance, dividing the dataset into separate training and testing sets. This approach offers a straightforward evaluation but can sometimes lead to variability in performance metrics depending on the data split.

To obtain a more representative and stable estimate of model performance, we further applied k-fold cross-validation [3]. This technique partitions the data into k equal-sized folds, iteratively using one fold as the test set and the remaining k-1 folds as the training set. This process is repeated k times, with each fold used exactly once as the test set. The average performance across all folds is considered a more reliable indicator of model efficacy due to its comprehensive utilization of the data for both training and testing. Employing k-fold cross-validation helps mitigate the limitations associated with the train-test split method, providing a robust framework for model evaluation. We used 5 folds for all our experiments.

3.2 Model Results

Our exploration began with the deployment of a Decision Tree as a baseline model. Decision Trees are particularly suited for baseline models in multi-class classification tasks due to their simplicity, interpretability, and the straightforward manner in which they handle categorical and numerical data. They work by iteratively splitting the data based on certain criteria to classify instances, making them a logical starting point for establishing a performance benchmark. For us, this model served as a first test to gauge the feasibility of our approach and the quality of our data.

Building on the foundation laid by the Decision Tree model, we advanced to a Random Forest Classifier. This ensemble method, which operates by constructing multiple decision trees at training time and outputting the class that is the mode of the classes of the individual trees, offers a significant step up in terms of prediction accuracy and overfitting control. Random Forests inherently manage biases and variances present in single decision trees, providing a more nuanced and robust model by aggregating the insights of numerous trees.

Finally, we experimented with XGBoost, an optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. XGBoost stands out for its ability to handle sparse data and its efficient implementation of the gradient boosting framework. It further improves model performance through advanced regularization (L1 and L2), which significantly reduces overfitting. XGBoost delivered the best results among the models we tested, combining high accuracy with the operational efficiency necessary for practical application.

To quantify the performance of our models, we analyzed their average cross-validation (CV) scores. Table 1 presents the average CV scores obtained from the Decision Tree, Random Forest Classifier, and XGBoost models. As shown, there is a progressive improvement in the average CV scores as we move from our baseline Decision Tree model to more sophisticated ones. Using a Random Forest

Classifier improves the average CV score to 86.37%. With XGBoost, which leverages state-of-the-art gradient boosting techniques, we achieve an average CV score of 92.04% on our dataset, signifying a substantial enhancement in model performance.

Table 1: Average CV Scores of the Used Models

Model	Average CV Score
Decision Tree	80.78%
Random Forest Classifier	86.37%
XGBoost	92.04%

In Figure 1, we present the precision scores by class for the best-performing model, XGBoost. We compare the precision scores across different classes to identify areas where the model excels and aspects that may require further refinement. While our model is able to predict the class label 353 with high precision, it shows lower precision for classes 357 and 351. Overall, the model performs well across most classes, as the precision scores for all classes are equal to or above 80%.

4 DISCUSSION

This section delves into the critical examination of our project, shedding light on the encountered limitations, proposing possible improvements, and outlining directions for future work. Our investigation into deploying machine learning models for multi-class classification in the context of WiFi signal fingerprinting has revealed valuable insights while also uncovering areas ripe for enhancement.

4.1 Limitations

Our approach demonstrated promising results, particularly in environments such as public buildings where WiFi networks are abundant, enhancing the model’s ability to accurately classify different areas. However, several limitations have emerged:

- **Platform Dependency:** The application’s current implementation is restricted to Android devices. This limitation stems from iOS’s network scan capabilities, which are less permissive compared to Android, hindering our app’s functionality on Apple devices.
- **Manual Room Addition:** The necessity to manually add new rooms into the system poses challenges in scalability and ease of use. This process can become cumbersome with the expansion of the application’s scope.
- **Scan Frequency:** Android devices limit network scans to one every 30 seconds. This restriction impacts the app’s responsiveness and real-time tracking capabilities, potentially affecting user experience.
- **Environmental Specificity:** The model’s effectiveness is closely tied to environments with a high density of WiFi networks. Its performance may vary in areas with fewer networks, affecting its applicability across different settings.

4.2 Possible Improvements

In light of the identified limitations, we propose several improvements to refine the application’s performance and user experience:

- **Data Gathering Enhancements:** Developing more sophisticated and automated methods for data collection could significantly reduce the manual effort involved and improve the model’s accuracy. A more uniform distribution of labels across different environments would also enhance the model’s generalizability.
- **Expansion of Training Data:** Collecting a larger and more diverse dataset could further improve the model’s robustness. Incorporating data from a wider range of environments would likely enhance its predictive capabilities across various settings.

4.3 Future Work

Building on our current findings and proposed improvements, we envision several avenues for future research and development:

- **Continued and Automated Data Collection:** Ongoing efforts to gather data will be crucial in enhancing the model’s accuracy and adaptability. Investigating methods to automate data collection and room addition would significantly streamline the application’s maintenance and scalability. This could involve leveraging user-generated data like crowd-sourced labeling.
- **Inclusion of Navigation or Markers:** Integrating navigation features or virtual markers within the app could greatly enhance its utility, providing users with real-time guidance based on their location.

5 CONCLUSION

IntelliTrack represents a practical approach to enhancing indoor navigation in environments well-equipped with 5G routers. Through the application of WiFi fingerprinting, our project has explored the feasibility of using existing WiFi infrastructure to assist in precise localization within buildings. This method, characterized by its reliance on machine learning algorithms to interpret WiFi signal data, offers a straightforward solution to some of the challenges faced by traditional GPS-based systems in indoor settings. Future directions for this work include refining the machine learning model to improve accuracy and exploring the integration of additional data sources to enhance the system’s robustness. Further research could also examine the scalability of the approach to different environments and its adaptability to varying WiFi densities. Additionally, we aim to build on the insights gained from IntelliTrack to address the identified limitations and to continue the exploration of accessible and efficient methods for indoor positioning. One of our goals is to develop this navigation to work with the room finder from LMU in every LMU building.

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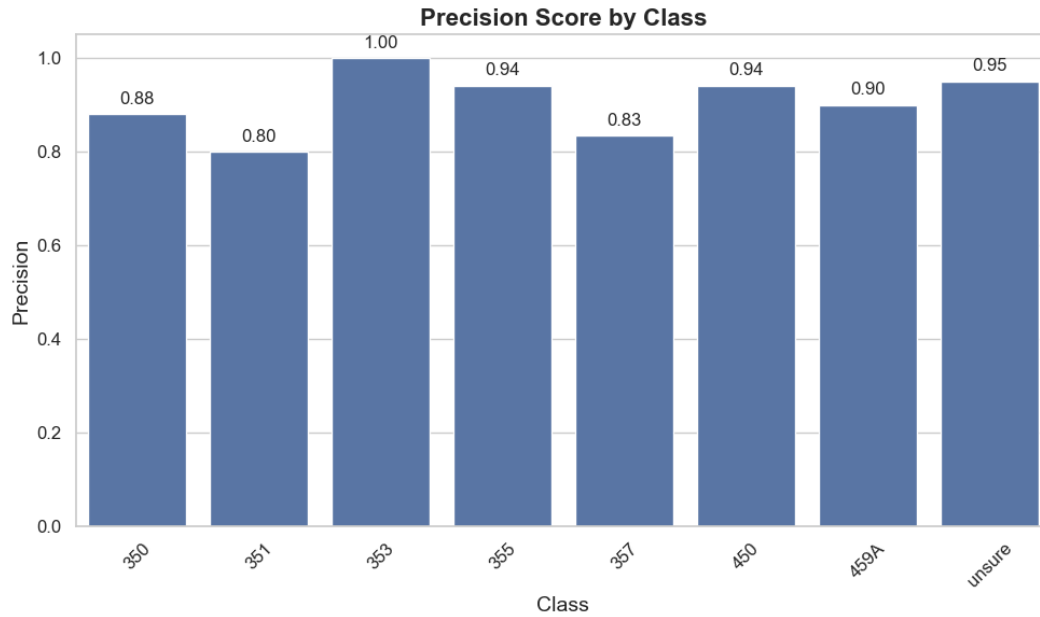


Figure 1: Precision scores by class with best XGBoost model

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